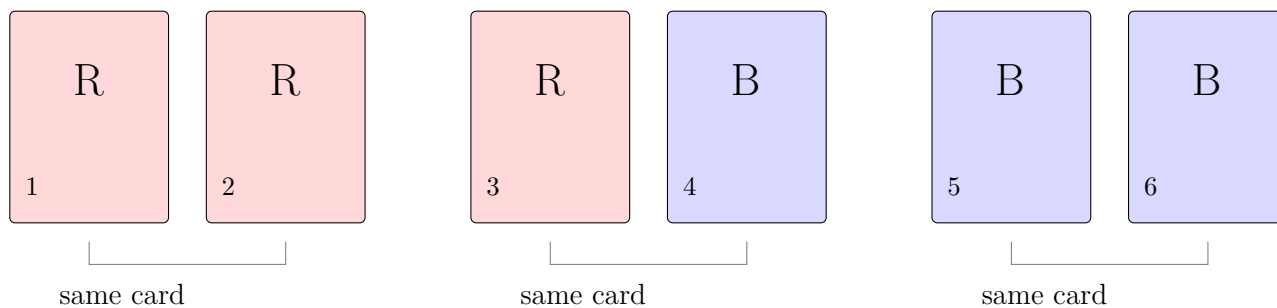
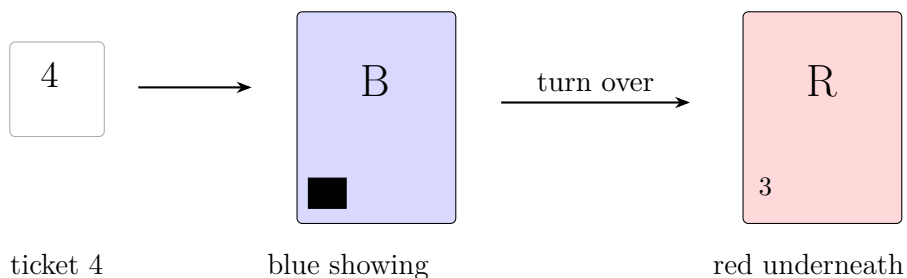


Each pair below is the two faces of one card. The small marks match the six tickets.

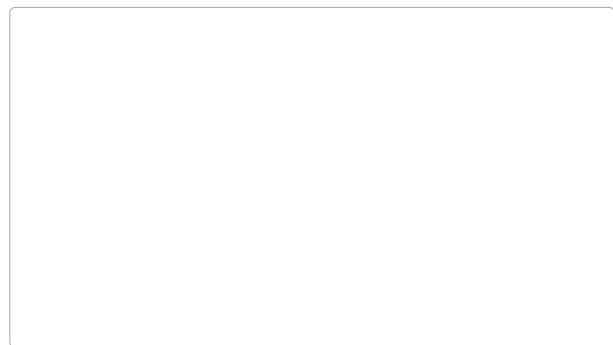


Use matching-size cards. Mix all six tickets and draw with equal chances. Behind a screen, choose the drawn face and set it upward. Keep the ticket secret and cover marks identically; reveal only the face color. Replace the ticket and mix each round.

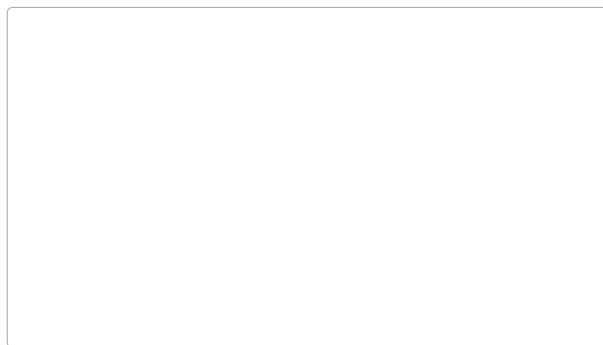


Problem 1: Sort every ticket by its hidden color. Then find two tickets that show the same color but hide different colors.

red underneath



blue underneath



Problem 2: Red is showing; which hidden color has more ways? Try removing one ticket, and find every removal that makes a tie.

red underneath

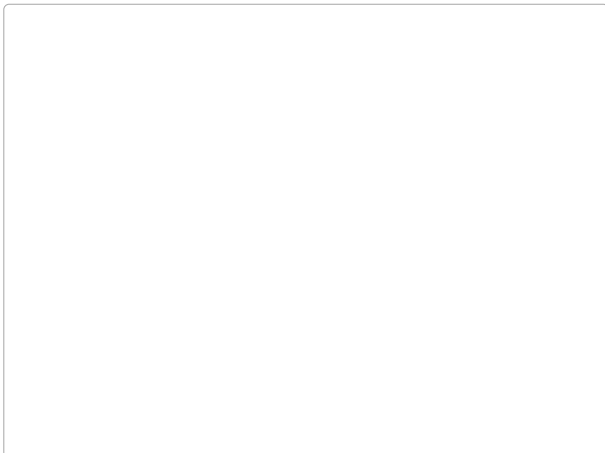


blue underneath

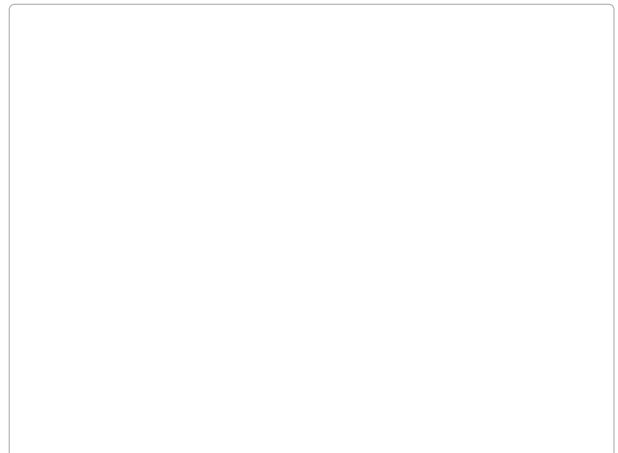


Problem 3: Put all six tickets back; now blue is showing. Find every one-ticket removal that gives a tie between the hidden colors.

red underneath



blue underneath



Use only tickets 1 and 3, with equal chances. Choose and orient behind the screen, keep the ticket secret, and show its red face with the mark covered. Replace and mix each round.



Problem 4: Add just one extra ticket to this chooser: 2, 4, 5, or 6, and show the face named by each draw. Among red-showing rounds, which hidden color has more ways before and after each addition?

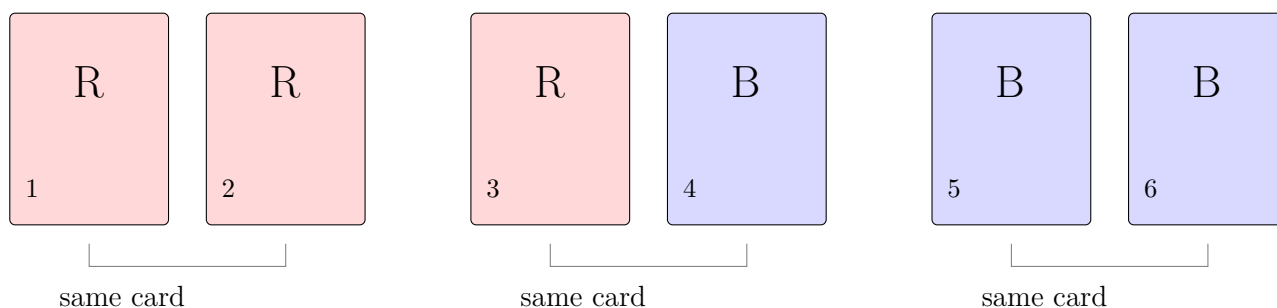
red underneath

blue underneath

Problem 5: Choose three tickets so red sometimes shows and then always hides blue. Find every choice that works.

Use all six face tickets again. When a card is removed, remove both its tickets.

Problem 6: Remove one card so red showing always hides red. Can you remove a different card so red showing always hides blue?



Problem 7: Find every two-ticket cup where red showing gives the hidden colors the same chance. Red must sometimes show.